Paper Title

Your Name
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1. Basic Probability Laws

These probability laws, such as the Chain Rule are fundamental. You can learn more about the link Chain Rule in this video.

1.1. Chain Rule

For two events A and B, the **Chain Rule** states

$$\mathbb{P}(A \cap B) = \mathbb{P}(B \mid A)\mathbb{P}(A).$$

1.2. PSRL

The PSRL algorithm [1] is shown in Algorithm 1.

Algorithm 1: Posterior Sampling for Reinforcement Learning (PSRL)

Input: Prior $P(\mathcal{M} \in \cdot)$
for $k \in [K]$ **do**
  Sample $M_k \sim \mathbb{P}(\mathcal{M} \in \cdot \mid H_k)$
  Obtain optimal policy $\pi^k = \pi^*_{M_k}$
  Execute $\pi^{\{(k)\}}$ and get trajectory τ_k
  Update history $H_{\{k+1\}} = H_k \cup \tau_k$
  Induce posterior $P\left(\mathcal{M} \in \cdot \mid H_{\{k+1\}}\right)$
end for

Now, consider an application of the Chain Rule, mentioned in Section 1.1:

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Bibliography

[1] M. Strens, "A Bayesian framework for reinforcement learning," in *ICML*, 2000, pp. 943–950.